Shenfu Xie 谢深甫

M.S. Student in Biomedical Engineering · Embedded Hardware · Medical Devices

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EDUCATION

City University of Hong Kong (Dongguan) *Sep 2025 – Present*
M.S. in Biomedical Engineering · Dongguan, China
Courses: Biomedical Instrumentation · Flexible Bioelectronics for Medical Applications

Greater Bay Area University *(Visiting Student)* *Sep 2025 – Present*
Medical Device R&D · Wearable Ultrasound System Development

Guangdong Medical University *Aug 2020 – Jul 2024*
B.S. in Biomedical Engineering · GPA: **84.7 / 100** · Rank: **TOP 1** · Guangdong, China
Key Courses: Signal & Linear System (91) · Microcomputer Principle & Interface (94) · MCU Principles (87) · Java Programming (99) · Biomedical Measurement & Sensors (95) · Medical Image Processing (100) · Medical Ultrasonic Diagnostic Equipment (95)

RESEARCH & PROFESSIONAL EXPERIENCE

Greater Bay Area University · Medical Device R&D *Sep 2025 – Present*
Research Member

- Participating in development of a **single-probe wearable ultrasound EMG system** and ultrasound excitation device
- Responsible for hardware assembly, component soldering, and signal acquisition core modules
- Advancing miniaturization and wearability to meet clinical application requirements

Shenzhen University General Hospital · Medical Engineering Dept. *Nov 2023 – Aug 2025*
Engineering Intern · Maintenance and Servicing of Medical Equipment

- Diagnosed persistent vs. intermittent faults; systematized circuit board inspection workflow: visual check → multimeter voltage test → cable/connector integrity → solder joint inspection → oscilloscope for complex faults
- Managed full **lifecycle maintenance** of medical devices; coordinated with third-party engineers for hardware upgrades
- Performed functional verification tests to confirm restored operation under practical conditions

Miniature Wireless Wearable Ultrasound Brain Stimulation System *Dec 2022 – May 2023*
Core Developer · Supervisor: Prof. Guofeng Li, Guangdong Medical University

- Designed 4-module wearable wireless ultrasound transmitter: **boost, power supply, remote control, and power amplification**
- Utilized DC-DC boost with PW6566 regulator; selected FNYK-01 (Tian R1A-M3-433 chip) for wireless signal reception
- Validated via rat ultrasound stimulation experiments; achieved significant **volume reduction and portability improvement**

Pulse Measurement System Based on STC89C51 MCU *Nov 2022 – May 2023*
Electronic Design Competition · Guangdong Medical University

- Built system around STC89C51 with DS18B20 temperature sensor, infrared contactless measurement module, and LCD display
- Diagnosed and resolved hardware faults using **logic analyzers and digital multimeters**; optimized PCB layout for reliability

Ultrasound Wave Analysis in Medical Imaging *Nov 2022 – May 2023*
Research Project · Guangdong Medical University

- Implemented **A-mode ultrasound imaging** via MATLAB Hilbert transform on RF data to extract signal envelope
- Generated B-mode images using threshold segmentation and morphological operations; measured blood vessel diameters

AWARDS & HONORS

Third Prize · 7th Electronic Design Competition, Guangdong Medical University	*2023*
Finalist Award · Guangdong Province "Hetai Cup" MCU Application Design Competition	*2023*
Academic Second Prize Scholarship (Rank TOP 1), Guangdong Medical University	*2023*
Outstanding Student Cadre Award, Guangdong Medical University	*2022 & 2021*
Academic Third Prize Scholarship, Guangdong Medical University	*2022 & 2021*
Certificate in Traditional Chinese Acupuncture (Intermediate Level)	*2022*

SKILLS

Programming	C, C++, Python, MATLAB, Java, Assembly Language
Hardware	Embedded Hardware Design, PCB Fabrication, Soldering, Prototype Assembly, Signal Acquisition
EDA / Tools	Altium Designer, Keil uVision, Tina-TI, EDA Software, MASM/DOSBox
Instruments	Oscilloscope, Digital Multimeter, Logic Analyzer
Languages	Chinese (Native) · English (IELTS 6.0, CET-6)
Other	Driver License C1 · TCM Acupuncture Certificate (Intermediate)